MIL-T-48319(PA)
30 September 1974
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

TRANSFORMER

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of transformer entitled as Transformer, Power which is used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8800544	- Carton, Packing, Ammunition for Transformer, Power, T1.
8808362	- Container, Packing, Ammunition for Domestic (CONUS) Shipments of Special Weapons Items, Packing and Marking.
8813567	- Transformer, Power.
EL-8842549	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The power transformer shall comply with all requirements specified on the drawing (dwg) 8813567, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Operational voltages. With an input voltage of 115 plus or minus (±) 5 volts root mean square (vrms), 50 ± 2 Hertz (Hz) or 60 ± 2 Hz or 400 ± 40 Hz applied to the primary winding (terminals 1 and 2), the secondary winding voltages shall be in accordance with Table I when the currents of all secondary windings have been adjusted to the values gived in Table I. The test shall be conducted as specified in 4.4.1.

Table I

Terminals	Second Voltage (rms)	Secondary Current Amperes (rms) Coil #2	Secondary Current Amperes (rms) Coil #1
3,5	41 ± 1	0.575 ± 0.05	0.0023 ± 0.001
3,4-4,5(coil 2)	20.5 ± 0.5	0.575 ± 0.05	0.0023 ± 0.0001
6-8	820 ± 16	0.575 ± 0.05	0.0023 ± 0.0001
6,7-7,8(coil 1)	410 ± 8	0.575 ± 0.05	0.0023 ± 0.0001

3.5 Dielectric strength. Each transformer shall be capable of withstanding the voltages in Table II without indication of dielectric failure. The tests shall be conducted as specified in 4.4.2.

Table II

From Terminal	To Terminal	AC Voltage (volts) rms
1	7	1500 ± 100
7	4	1500 ± 100
7	Case	1500 ± 100
4	Case	1500 ± 100

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3.6 Insulation resistance. Insulation resistance shall be 20 megohms or greater when tested at 500 volts ± 50 volts direct current (vdc) between the terminals specified in Table III. Testing shall be as specified in 4.4.3.

Table III

From Terminals	To Terminals	Impressed Voltage	Terminals Grounded to Case	Resistance in Megohms, min
1	Case	500 ± 50 vdc	4,7	20
1	4	500 ± 50 vdc	7	20
1	7	500 ± 50 vdc	4	20

3.7 Induced voltage. A voltage of 230 ± 20 vrms at 400 ± 20 Hz shall be applied to terminals 1 and 2 for 18 seconds maximum (max) without the occurence of any abrupt current changes in the nominal value of the primary current. Testing will be as specified in 4.4.4.

3.8 Environmental testing. The power transformer shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The power transformer shall also comply with 3.4 before and after each environmental test. Testing shall be as specified in 4.4.5. Tests shall be conducted in the order specified.

3.8.1 High temperature. The transformer shall withstand subjection to a temperature of 160 ± 5°F without damage and shall meet the requirements of 3.4 at a temperature of 160 ± 5°F when tested as specified in 4.4.5.1.

3.8.2 Low temperature. The transformer shall withstand subjection to a temperature of -80 ± 5°F without damage and shall meet the requirements of 3.4 while at -65 ± 5°F when tested as specified in 4.4.5.2.

3.8.3 Sealing. The transformer shall be submerged in water 140 ± 15°F for 2 minutes minimum without evidence of bubble emission. Testing will be as specified in 4.4.5.3.

3.8.4 Vibration. The transformer shall withstand, without damage, vibration in three mutually perpendicular axes at the frequencies, amplitudes, temperatures and times specified in 4.4.5.4. Upon completion of this test the transformer shall meet the requirements of 3.4.

Table IV

Frequency Range Hertz (Hz)	Double Amplitude or Peak g Level (± 5 percent)	Cycle Time to Cover Frequency Range in Minutes	Total Vibration Time min - Minutes 80 + 20°F	160 + 5°F	-65 + 5°F
10-500-10	0.036 or 10 g	5 ± 5 percent	30	10	10

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3.8.5 Shock. The transformer shall withstand, without damage, three shocks in each of its three major and mutually perpendicular axes at the amplitude and temperatures and for the time specified in Table V. Testing shall be as specified in 4.4.5.5. Upon completion of this test the transformer shall meet the requirements of 3.4.

Table V

Peak Shock Amplitude	Time Duration Half Amplitude	Temperature
20 ± 2 g	10 ± 3 milliseconds	-65 ± 5°F
20 ± 2 g	10 ± 3 milliseconds	80 ± 20°F
20 ± 2 g	10 ± 3 milliseconds	160 ± 5°F

3.9 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 5 complete assemblies (see dwg 8813567) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VI. Tests shall be conducted in order specified in Table VI.

Table VI

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Operational voltages (see 3.4)		X
3. Dielectric strength (see 3.5)		X
4. Insulation resistance (see 3.6)		X
*5. High temperature (see 3.8.1)		X
*6. Low temperature (see 3.8.2)		X
*7. Vibration (see 3.8.4)	X	
*8. Shock (see 3.8.5)	X	
9. Sealing (see 3.8.3)		X
10. Induced voltage (see 3.7)		X

*Note: Sequence of these tests is optional.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VII and Table VIII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table VII
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VIII
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Transformer, before sealing (see dwg 8813567).

Categories		Characteristics	Method of Inspection
Critical:	None defined.		
Major:			
	101.	Improper soldering (see 3.9)	Visual
	102.	Any component missing, loose or damaged	Visual/Manual
	103.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:			
	201.	Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.2 Transformer (see dwg 8813567).

Categories		Characteristics	Method of Inspection
Critical:	None defined.		
Major:			
	101.	Location of mounting studs	Gage
	102.	Improper size of mounting studs	Gage
	103.	Broken or cracked insulation around terminals	Visual
	104.	Broken or distorted terminals	Visual
	105.	Mounting studs damaged	Visual

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106. Outer case bent, punctured or distorted.....Visual
107. Improper soldering (see 3.9).....Visual
108. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Marking missing or incorrect.....Visual
202. Evidence of poor workmanship (see 3.10).....Visual

4.3.2.1.3 Carton, packing, ammunition for transformer, power, T1 (see dwg 8800544).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Wrong size bag	Visual
	102. Bag not sealed	Visual
	103. Air in bag not minimized before sealing	Visual
	104. Incorrect or insufficient cushioning material	Visual
	105. Evidence of poor workmanship (see 3.10)	Visual
	106. Marking missing, incorrect or mislocated	Visual
	107. Packing material visibly wet	Visual
	108. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.4 Container, packing, ammunition for domestic, (CONUS) shipments of special weapons items, packing and marking (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect or insufficient cushioning material	Visual

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102. Marking missing, incorrect or mislocated.....Visual
103. Packing material visibly wet.....Visual
104. Evidence of poor workmanship (see 3.10).....Visual
105. Contents loose.....Manual
106. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor: None defined.

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Operational voltage (60 ± 2 Hz only)	Major
*b.	Dielectric strength	Major
*c.	Insulation resistance	Major
d.	Sealing	Major
e.	High temperature	Major
f.	Low temperature	Major
g.	Induced voltage	Major

*Note: Do these tests in the order specified.

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Operational voltage, dielectric strength, insulation resistance and sealing. Each transformer in the lot shall be tested in accordance with 4.4.1, 4.4.2, 4.4.3 and 4.4.5.3. If any item fails to comply with 3.4, 3.5, 3.6 and 3.8.3, the item shall be rejected and removed from the lot.

4.3.3.2 High temperature, low temperature and induced voltage. A sample shall be selected at random in accordance with Table IX. The sample shall be tested in accordance with 4.4.5.1, 4.4.5.2 and 4.4.4. If any item fails to comply with 3.8.1, 3.8.2 and 3.7, the lot shall be rejected.

Table IX
Sample sizes for non-destructive testing

Lot Size	Sample Size
2-180	2
181-500	3
501-800	5
801-1300	7

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842549). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified all tests shall be performed at 80 + 20°F.

4.4.1 Operational voltages. Use a suitable alternating current (ac) voltage and frequency source to apply the specified voltages to the primary winding. Use a variable resistance fixture to set the secondary currents to the limits specified in Table I. Using a suitable ac voltage measuring instrument, measure the voltages between the respective terminals of Table I. When measuring the voltage across terminals 6-7-8, an instrument having a minimum input impedance of 10 megohms must be used to avoid loading effects.

4.4.2 Dielectric strength. A suitable dielectric test instrument shall be used to apply the specified voltage to the test points specified in Table II. The test instrument shall be current limited to 5 milliamperes to protect human life.

4.4.3 Insulation resistance. An insulation resistance tester capable of supplying the required voltage and with which a resistance in megohms can be obtained, shall be used to measure the resistance. The current shall be limited to a maximum of 5 milliamperes to protect human life.

4.4.4 Induced voltage. A suitable ac current measuring device shall be used to detect abrupt changes in the nominal value of the primary current when a voltage of 230 ± 20 volts ac, rms, 400 Hz is applied to terminals 1 and 2.

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4.4.5 Environmental testing (see 3.8).

4.4.5.1 High temperature. The transformer shall be placed within a chamber and the internal temperature of the chamber raised to 160 + 5°F. The transformer shall be maintained at 160 + 5°F for a period of 2 hours minimum or until stabilization (see 6.4), whichever is longer. At the conclusion of the exposure period and while still at this temperature, the transformer shall be tested as specified in 4.4.1.

4.4.5.2 Low temperature. The transformer shall be placed within a chamber and the chamber cooled to, and maintained at, a temperature of -80 + 5°F for a period of 4 hours. The temperature of the chamber shall then be gradually raised to -65 + 5°F and maintained for an additional 2 hour period or until stabilization (see 6.4), whichever is longer. At the conclusion of this exposure period and while at this temperature, the transformer shall be tested as specified in 4.4.1.

4.4.5.3 Sealing. The transformer at 80 + 20°F shall be submerged in water at 140 + 15°F for 2 minutes minimum. Leaks will be evidenced by the emission of bubbles at any point. Small bubbles formed by the release of entrapped surface air shall not be interpreted as leaks.

4.4.5.4 Transportation vibration. A mounting fixture capable of holding the assembly in the three major axes shall be fastened securely to a suitable vibration machine. The assembly shall be vibrated at the frequencies, amplitudes, temperatures, and for the times specified in Table IV in each of the three mutually perpendicular axes. At the conclusion of this test, the transformer shall be tested as specified in 4.4.1.

4.4.5.5 Transportation shock. The transformer shall be rigidly mounted to a fixture and the fixture mounted to a suitable shock producing device. The shock producing device shall be placed in a suitable test chamber and the chamber's temperature shall be lowered to -65 + 5°F and stabilized (see 6.4). The transformer shall then be shocked once in each of its three major and mutually perpendicular axes in accordance with Table V. This procedure shall be repeated at temperatures of 80 + 20°F and 160 + 5°F. Upon completion of this test the transformer shall be tested in accordance with 4.4.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The power transformer shall be packaged in accordance with dwg 8800544 using the specified moisture barrier.

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5.1.2 Level C. The power transformer shall be packaged in accordance with dwg 8800544 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The power transformer shall be packed in accordance with dwg 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwgs 8800544 and 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1690 dated 1 August 1961.

Custodian:
Army - PA

Preparing activity:
Army - PA

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